NAND2Tetris / Hardware Simulator / Register.hdl

HDL | Builtin | Project 3 | Register

```
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/Register.hdl
/**
 * 16-bit register:
 * If load is asserted, the register's value is set to in;
 * Otherwise, the register maintains its current value:
 * if (load(t)) out(t+1) = int(t), else out(t+1) = out(t)
 */
CHIP Register {
    IN in[16], load;
    OUT out[16];

    PARTS:
```

Chip Register | Eval | Reset | Clock: 74

Input pins

in	0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1	dec
load	1	

Output pins

out	0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1	dec

Test | **Steps: 151 Outputs: 149** Slow — Fast

Test Script | Compare File | Output File | Diff Table

```
output;

set load 1,
tick,
output;

tock,
```

NAND2Tetris / Hardware Simulator / RAM8.hdl

HDL Builtin | Project 3 | RAM8

```
/**
 * If load is asserted, the value of the register selected by
 * address is set to in; Otherwise, the value does not change.
 * The value of the selected register is emitted by out.
 */
CHIP RAM8 {
    IN in[16], load, address[3];
    OUT out[16];

    PARTS:
    DMux8Way(in =load,sel=address,a=l0,b=l1,c=l2,d=l3,e=l4,f=l5,g=l6,h=l7);
    Register(in=in,load=l0,out=r0);
    Register(in=in,load=l1,out=r1);
    Register(in=in,load=l2,out=r2);
    Register(in=in,load=l3,out=r3);
```

Chip RAM8 | Eval | Reset | Clock: 46

Input pins

in	0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
load	0
address	1 1 1 dec

Output pins

out	0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec

Internal pins

l0	0
l1	0

Test | Steps: 160 Outputs: 173 Slow — Fast

Test Script | Compare File | Output File | Diff Table

```
output;
set address 6,
eval,
output;
set address 7,
eval,
output;
```

NAND2Tetris / Hardware Simulator / RAM64.hdl

HDL | Builtin | Project 3 | RAM64

```
1  // This file is part of www.nand2tetris.org
2  // and the book "The Elements of Computing Systems"
3  // by Nisan and Schocken, MIT Press.
4  // File name: projects/3/a/RAM64.hdl
5
6  /**
7   * Memory of sixty four 16-bit registers.
8   * If load is asserted, the value of the register selected by
9   * address is set to in; Otherwise, the value does not change.
10  * The value of the selected register is emitted by out.
11  */
12
13 CHIP RAM64 {
14     IN in[16], load, address[6];
15     OUT out[16];
```

Chip RAM64 | Eval | Reset | Clock: 81

Input pins

in	0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1	dec
load	0	
address	1 1 1 1 0 1	dec

Output pins

out	0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1	dec

Internal pins

l0	0
l1	0

Test | **Steps: 292 Outputs: 320** Slow — Fast

Test Script | Compare File | Output File | Diff Table

```
1018 eval,
1019 output;
1020 set address %B110101,
1021 eval,
1022 output;
1023 set address %B111101,
1024 eval,
```

NAND2Tetris / Hardware Simulator / RAM512.hdl

HDL Builtin | Project 3 | RAM512

```
1  CHIP RAM512 {
2      IN in[16], load, address[9];
3      OUT out[16];
4
5      PARTS:
6
7      DMux8Way(
8          in=load,
9          sel=address[0..2],
10         a=load0,
11         b=load1,
12         c=load2,
13         d=load3,
14         e=load4,
15         f=load5,
```

Chip RAM512 | Eval | Reset | Clock: 81

Input pins

in	0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1	dec
load	0	
address	1 1 1 1 0 1 0 1 0	dec

Output pins

out	0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1	dec

Internal pins

load0	0
load1	0

Test | Steps: 292 Outputs: 320 Slow — Fast

Test Script | Compare File | Output File | Diff Table

```
1019 output;
1020 set address %B110101010,
1021 eval,
1022 output;
1023 set address %B111101010,
1024 eval,
1025 output;
```

NAND2Tetris / Hardware Simulator / RAM4K.hdl

HDL Builtin | Project 3 | RAM4K

```
1  CHIP RAM4K {
2      IN in[16], load, address[12];
3      OUT out[16];
4
5      PARTS:
6
7
8      DMux8Way(
9          in=load,
10         sel=address[0..2],
11         a=load0,
12         b=load1,
13         c=load2,
14         d=load3,
15         e=load4,
16         f=load5,
```

Chip RAM4K | Eval | Reset | Clock: 81

Input pins

in	0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1	dec
load	0	
address	1 1 1 1 0 1 0 1 0 1 0 1	dec

Output pins

out	0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1	dec

Internal pins

load0	0
load1	0

Test | **Steps: 292 Outputs: 320** | Slow | Fast

Test Script | Compare File | Output File | Diff Table

```
1018 eval,
1019 output;
1020 set address %B110101010101,
1021 eval,
1022 output;
1023 set address %B111101010101,
1024 eval,
```

NAND2Tetris / Hardware Simulator / RAM16K.hdl

HDL | Builtin | Project 3 | RAM16K

```
1  CHIP RAM16K {
2      IN in[16], load, address[14];
3      OUT out[16];
4
5      PARTS:
6
7
8      DMux4Way(
9          in=load,
10         sel=address[12..13],
11         a=load0,
12         b=load1,
13         c=load2,
14         d=load3
15     );
```

Chip RAM16K | Eval | Reset | Clock: 81

Input pins

in	0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
load	0
address	1 1 1 1 0 1 0 1 0 1 0 1 0 1 dec

Output pins

out	0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec

Internal pins

load0	0
load1	0

Test | **Steps: 292 Outputs: 320** Slow — Fast

Test Script | Compare File | Output File | Diff Table

```
1018 eval,
1019 output;
1020 set address %B11010101010101,
1021 eval,
1022 output;
1023 set address %B11110101010101,
1024 eval,
```

NAND2Tetris / Hardware Simulator / PC.hdl

HDL | Builtin | Project 3 | PC

```
1  CHIP PC {
2      IN in[16], reset, load, inc;
3      OUT out[16];
4
5      PARTS:
6      Register(in=next, load=true, out=current);
7
8      Inc16(in=current, out=incValue);
9
10     Mux16(a=current, b=incValue, sel=inc, out=incOrHold);
11
12     Mux16(a=incOrHold, b=in, sel=load, out=loadOrInc);
13
14     Mux16(a=loadOrInc, b=false, sel=reset, out=next);
15
```

Chip PC | Eval | Reset | Clock: 15+

Input pins

in	1 0 0 1 0 0 1 0 0 1 0 0 1 0 0 1
reset	1
load	0
inc	0

Output pins

out	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Internal pins

next	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Test | **Steps: 33 Outputs: 31** Slow — Fast

Test Script | Compare File | Output File | Diff Table

```
117 set in 22222,
118 set reset 1,
119 set inc 0,
120 tick,
121 output;
122
123 tock,
```

```
// File name: projects/3/a/Bit.hdl
/**
 * 1-bit register:
 * If load is asserted, the register's value is set to in;
 * Otherwise, the register maintains its current value:
 * if (load(t)) out(t+1) = in(t), else out(t+1) = out(t)
 */
CHIP Bit {
    IN in, load;
    OUT out;

    PARTS:
    Mux(a=outLoop, b=in, sel=load, out=muxOut);
    DFF(in=muxOut, out=outLoop, out=out);
}
```

Input pins

in	1
load	1

Output pins

out	0

Internal pins

outLoop	0
muxOut	1

Test **Steps: 217 Outputs: 215** Slow Fast

Test Script | Compare File | Output File | Diff Table

```
set in 1,
set load 0,
tick,
output;

tock,
```